МЕЖДУНАРОДНЫЙ УНИВЕРСИТЕТ INTERNATIONAL UNIVERSITY
ХАЛЫҚАРАЛЫҚ УНИВЕРСИТЕТІ

Faculty of Computer Technology and Cyber Security

Laboratory work №6
By discipline: “Fundamentals of Logic Design”
On the topic: “Candlestick design”

Done by: student of group IT2-2101
Diana Sarsembayeva
Checked by: Senior-lecturer
Iskakov E.A.

Almaty, 2024

Introduction

The purpose of this project is to create a candlestick following the instructions in Building Models: Revolves and Sweeps. Creating a candlestick will not only allow you to learn how to use the program tools to create complex shapes, but will also provide practical benefits in the form of a finished product that can be used in everyday life.

Workflow

Model creation begins with a sketch of a two-dimensional model. The sketch starts on the front plane. Starting from the center of the plane, a line upwards of 220 mm is drawn, and lines of 20 and 90 mm are drawn at the top and bottom, respectively.

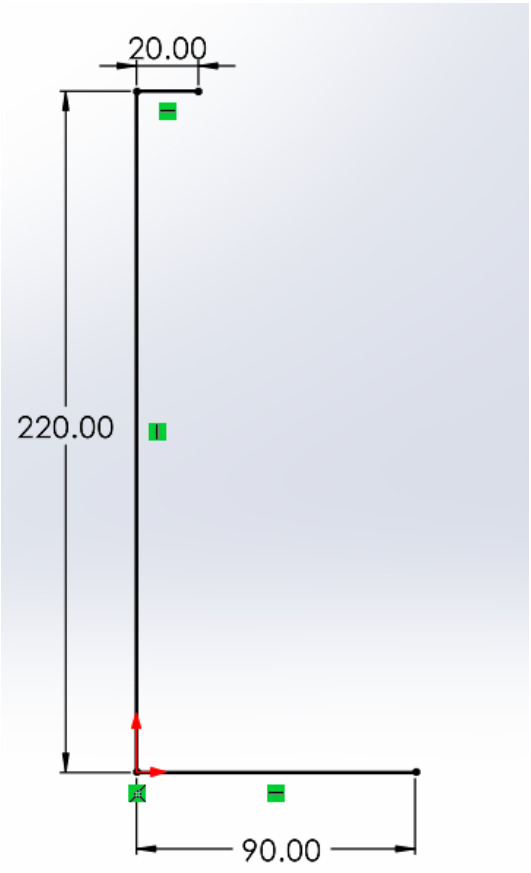


Figure 1.1 – Sketching a Revolve Profile

The next step is to finish creating half of the two-dimensional sketch of the candlestick. Let's start with the top two semicircles. The first one is in the sketch header with a radius of 10mm, the second one with a radius of 20mm and located 40mm below the first one. Next, there are also two connected semicircles. They are located 40mm below. The first has a radius of 60mm and the second has a radius of 12.50mm.

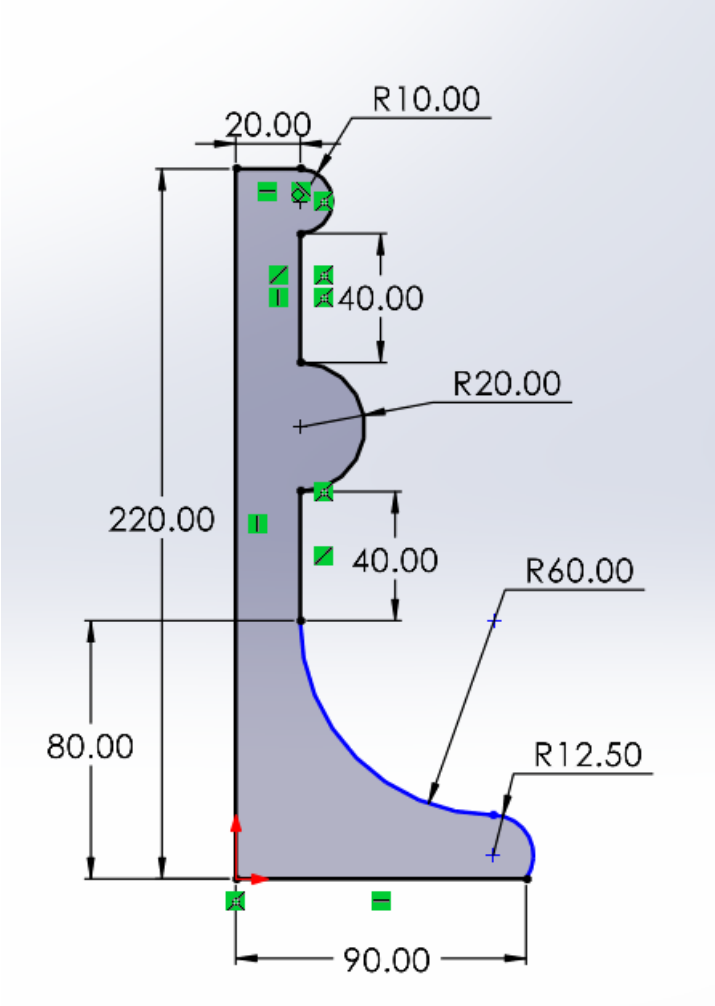


Figure 1.2 – Completing the Revolve Profile

In order to create a three-dimensional model, we use the 'Boss/Base Revolve' tool, in which the profile is rotated around the centerline, in our case the centerline will be the left line 220 mm long.

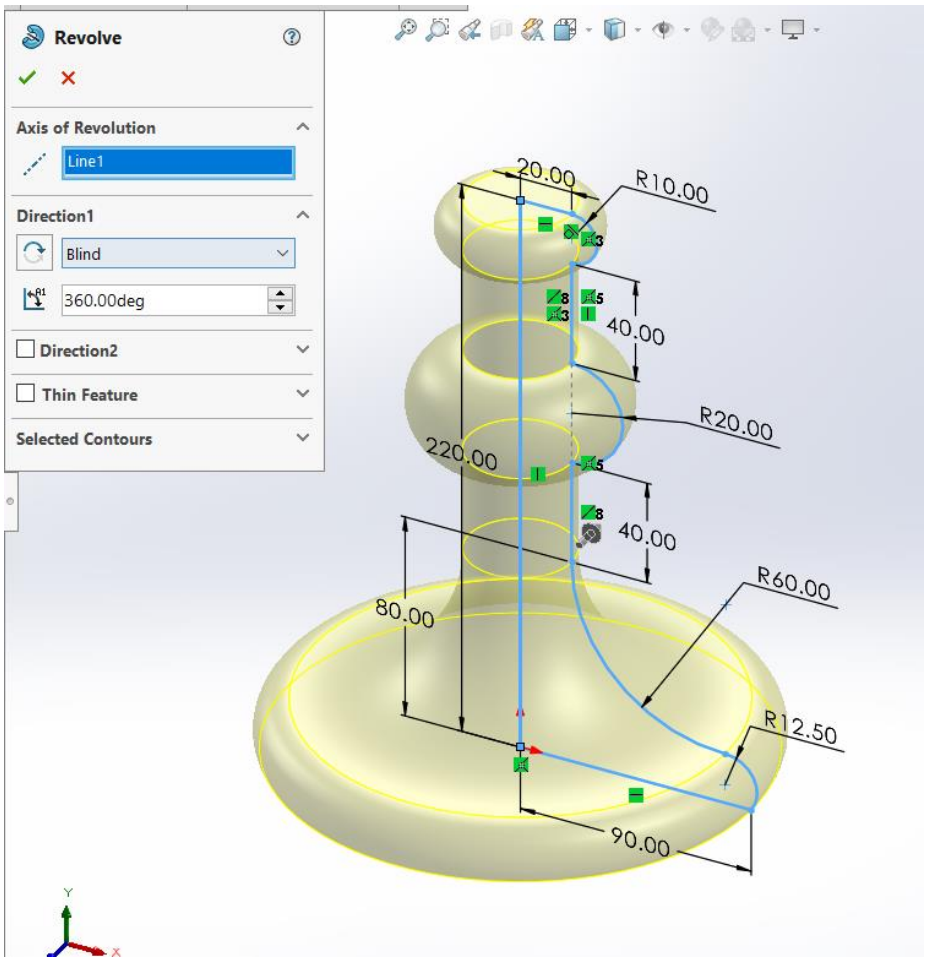


Figure 1.3 – Creating the Revolve Feature

Create a candle handle using sweep. First sketch the sweep path on the front plane. The handle will be located 10mm above the base. The handle itself consists of a 60mm line, two semi-circles with a radius of 150mm and 20mm.

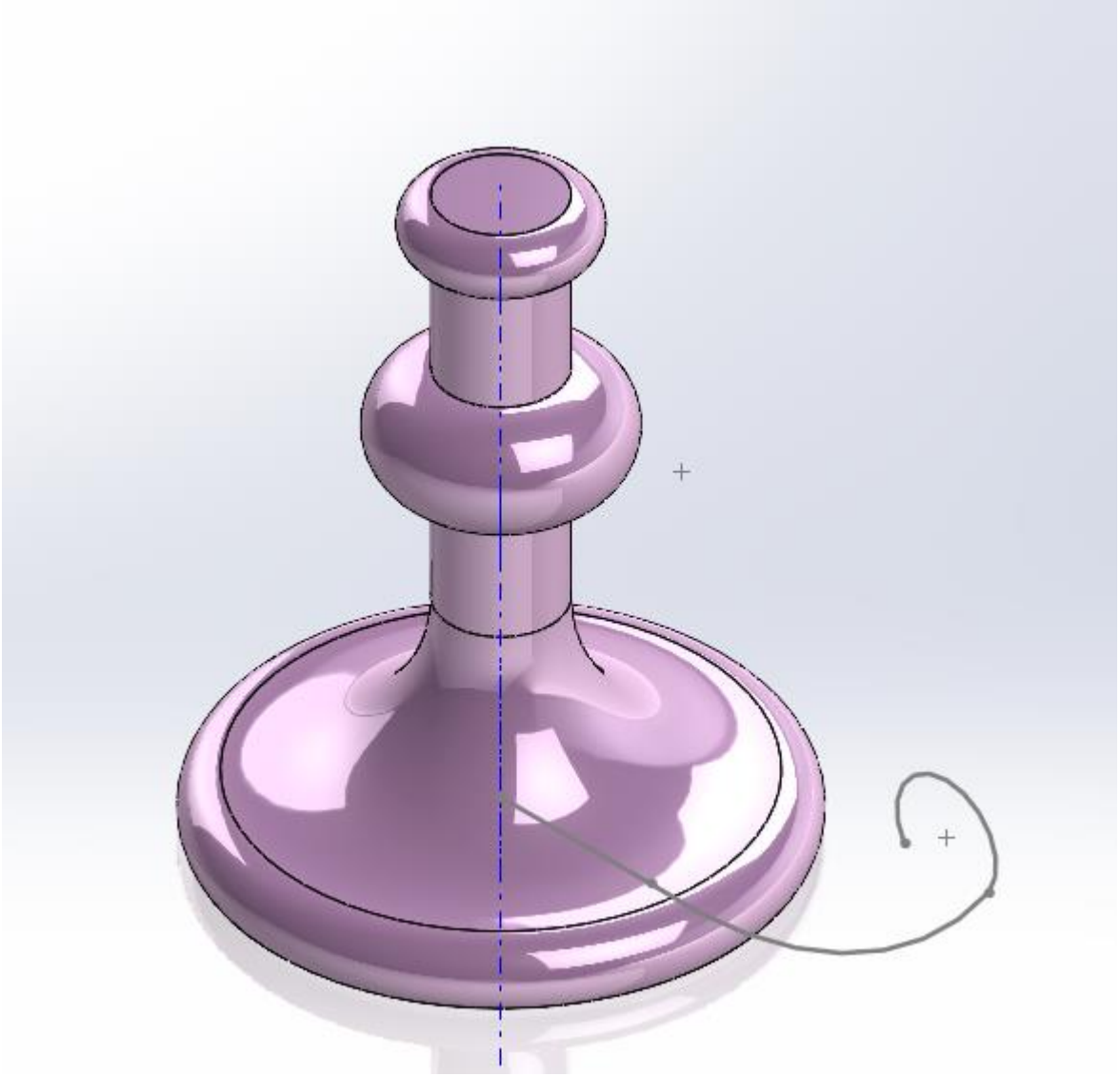


Figure 1.4 – Completing the Sweep Path

Our next step is to finish the Sketch of the sweep section. This is done by creating an oval with dimensions of 35mm long and 10mm wide. The oval will be placed in the center of the candleholder and connected to the beginning of the handle.

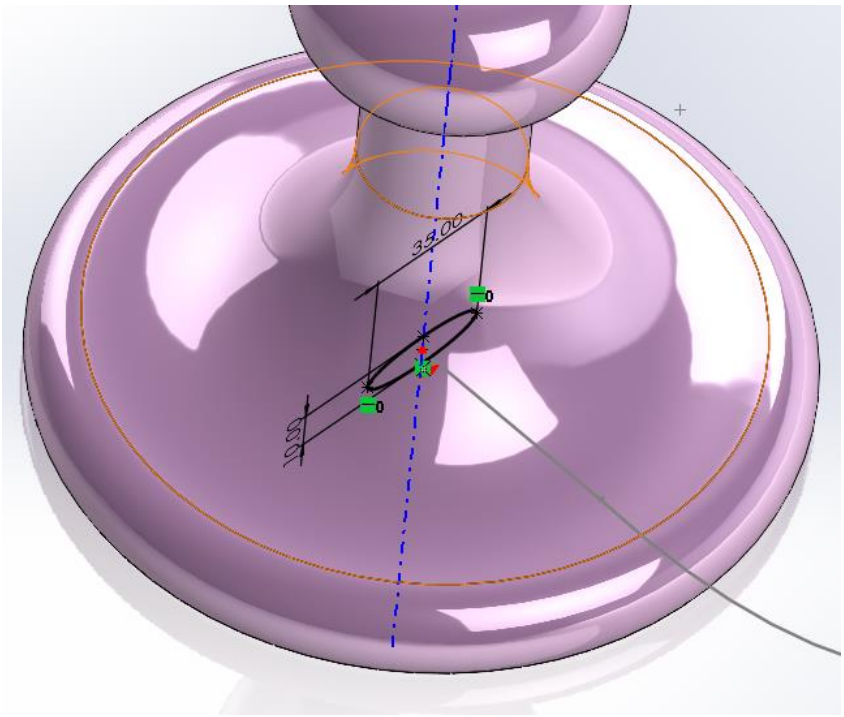


Figure 1.5 – Completing the Sweep Section Sketch

Now you combine the sweep path and sweep section sketches to create the sweep.

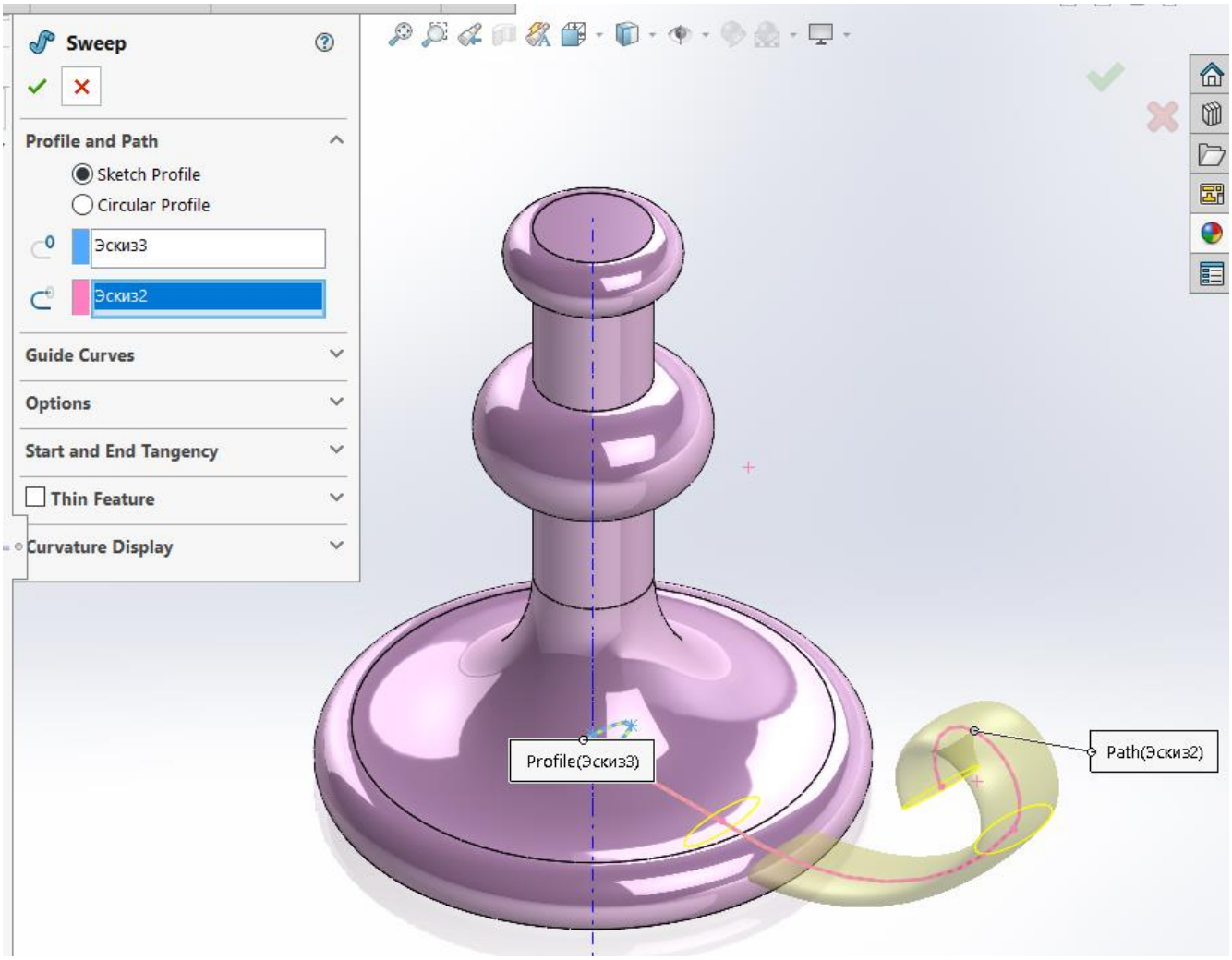


Figure 1.5 – Creating the Sweep

The last step is to create the cut out for the candle. Sketch and dimension a circle that is located on the top face of the rotating base. The diameter of the circle is 30mm. Then, use the "Extruded Cut" tool, which has Depth to 25, Click Draft On, and set Draft Angle to 15.



Figure 1.5 – Final result

Conclusion

This lesson is a great opportunity to learn basic modeling techniques in the program, specifically creating samples by rotating and slicing. Each step of creating a candlestick was accomplished by following the lesson instructions and using the various functions and tools. This experience is useful not only for understanding how the program works, but also for developing creative skills in design and modeling. Practicing such lessons helps to deepen knowledge and confidence in working with the program.